FHIA-03 — Building a Trustworthy Longitudinal Patient History

From Healthcare Data to Clinical Knowledge

DOUGLAS EBEL

AUGUST 13, 2026

FUTURE HEALTHCARE INFORMATION ARCHITECTURE (FHIA) SERIES

A national longitudinal patient history requires more than exchanging healthcare records. This paper describes the information needed to create that history and how structured data and clinical documents can be transformed into evidence-linked, longitudinal clinical knowledge while preserving conflicting information, provenance, and temporal context.

Contents

Introduction

The previous paper described an important limitation of today's healthcare interoperability architecture. Standards such as TEFCA support retrieving information from other healthcare organizations when it is requested, but a longitudinal patient history must also be maintained as new clinical events occur and existing information is corrected or changed. That requires a mechanism for participating systems to identify and communicate those changes rather than waiting for another system to request the information.

If that information can be continuously assembled, the next question is what the resulting longitudinal patient history would contain and how it could improve healthcare decisions and reduce costs.

Technical teams designing information systems commonly use **logical data models** and **physical data models** to define information in considerable detail, including individual data elements, relationships, keys, and how the data will ultimately be stored. That level of detail is not necessary to understand the proposed architecture.

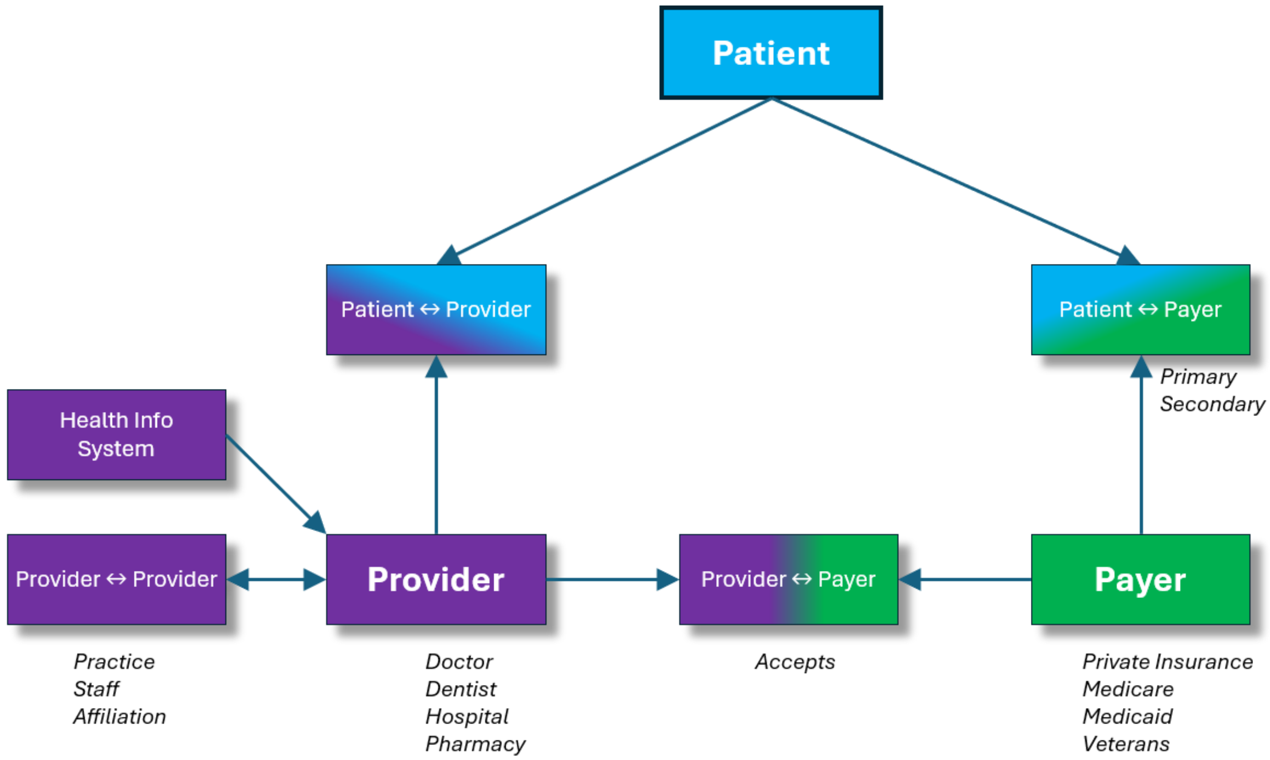
Instead, this paper uses a **conceptual information model**. It identifies the major types of information that would be available in a longitudinal patient history and the important relationships among them. For example, a patient may have multiple medical conditions; those conditions may be assessed during multiple clinical encounters; an encounter may produce clinical notes, observations, laboratory results, imaging studies, procedures, or diagnoses; and procedures may subsequently result in claims and payments.

Understanding these relationships helps explain what becomes possible when information that is currently fragmented across healthcare organizations is assembled into a longitudinal history. It also helps distinguish information that can be obtained directly from existing healthcare systems from clinical knowledge that must be extracted, reconciled, and maintained over time.

The following diagrams progressively build the conceptual model, beginning with the patient, provider, and payer relationships and then adding the patient, clinical, and administrative information needed to create a useful longitudinal history.

The Healthcare Participants

At the highest level, healthcare involves three principal participants: **patients, providers, and payers**.



A patient may receive care from many providers over time, while each provider serves many patients. The **Patient↔Provider** relationship represents this many-to-many relationship and provides a way to associate a patient with the providers and healthcare organizations involved in their care.

Similarly, a patient may be covered by different payers over time, and sometimes by more than one payer at the same time. **Patient↔Payer** represents this relationship and allows coverage to be associated with the periods during which it applies.

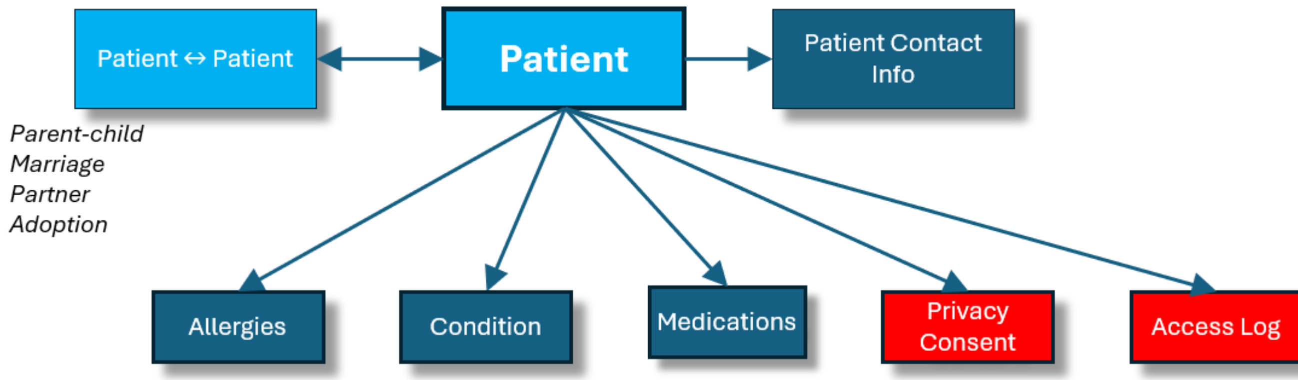
Providers also have relationships with other providers and healthcare organizations. A physician may belong to a medical practice, a practice may be affiliated with a hospital, and hospitals, laboratories, imaging centers, and pharmacies may participate in the delivery of care. These relationships are represented by **Provider↔Provider**.

Finally, **Provider↔Payer** represents the relationships between providers and the payers that accept their insurance plans.

Existing Healthcare Information Systems remain associated with the providers that use them. They continue to support the operational functions of those organizations while supplying information needed to build and maintain the longitudinal patient history.

The Patient

The longitudinal history is organized around the patient rather than around any individual healthcare organization.



Basic demographic and contact information identifies the patient, but several other types of information need to follow the patient throughout their lifetime, independent of where care is received.

Patient ↔ Patient represents relationships with other patients, such as parent-child, marriage or partnership, and adoption. These relationships may provide clinically relevant family context while also supporting situations in which one individual is authorized to act on behalf of another.

Allergies identify substances that may cause an adverse reaction and need to be available whenever treatment decisions are made.

Conditions represent both current and historical health conditions. Some may be established through clinical care, while others may initially be known only because the patient reported them as part of their medical history.

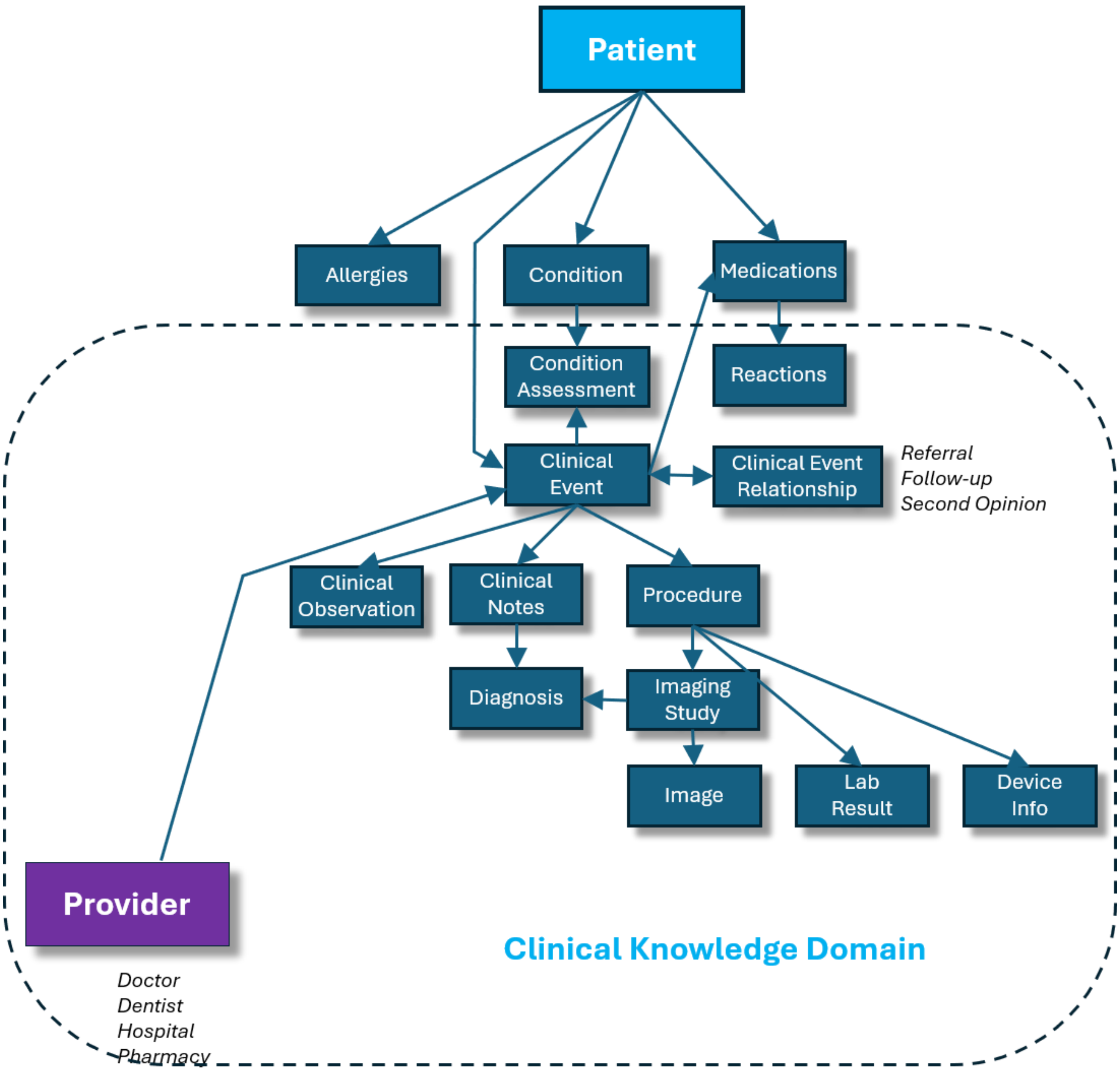
Medications provide a longitudinal record of medications the patient is taking or has taken. This includes medications prescribed by providers as well as over-the-counter medications reported by the patient. Maintaining medication information at the patient level allows it to persist independently of the individual clinical events in which medications may be prescribed, changed, discontinued, or reviewed.

Privacy Consent records the patient's choices governing access to their healthcare information, while the **Access Log** provides a history of who accessed that information, when it was accessed, and for what purpose.

Together, these elements establish information that belongs to the patient's longitudinal history rather than to any single provider, payer, or clinical encounter.

The Clinical Knowledge Domain

Most healthcare information is created during interactions between patients and providers. In this conceptual model, each of these encounters is represented as a **Clinical Event**. A routine physical, an emergency room visit, a specialist consultation, a hospital admission, labs, or a telehealth appointment are all examples of clinical events.



A clinical event may exist without being associated with a particular medical condition. A routine physical is one example. More commonly, however, one or more of the patient's conditions are evaluated during an encounter. Because a patient may have many conditions addressed during one clinical event, and a condition may be evaluated during many clinical events over time, **Condition Assessment** represents the relationship between conditions and the clinical events in which they are evaluated.

Clinical events produce much of the information traditionally maintained by Healthcare Information Systems. This may include **Clinical Notes, Clinical Observations, Diagnoses, Procedures, Laboratory Results, Imaging Studies, Images, and information about implanted Devices**. Providers may also prescribe, change, discontinue, or review medications during an encounter.

Not all of this information is equally structured. Laboratory results, medication orders, procedure codes, and many clinical observations may already be represented using defined fields and healthcare standards. Clinical notes and imaging reports contain additional knowledge expressed as narrative text. Clinical Knowledge Extraction must identify clinically significant information within those documents while retaining the original source material and its provenance.

Conditions and Condition Assessments add longitudinal context to this source information. A diagnosis recorded during one encounter may identify a new condition, confirm an existing condition, modify an earlier assessment, or conflict with information recorded elsewhere. Rather than treating every diagnosis or mention in a clinical note as an independent fact, the longitudinal architecture must reconcile this information with what is already known about the patient.

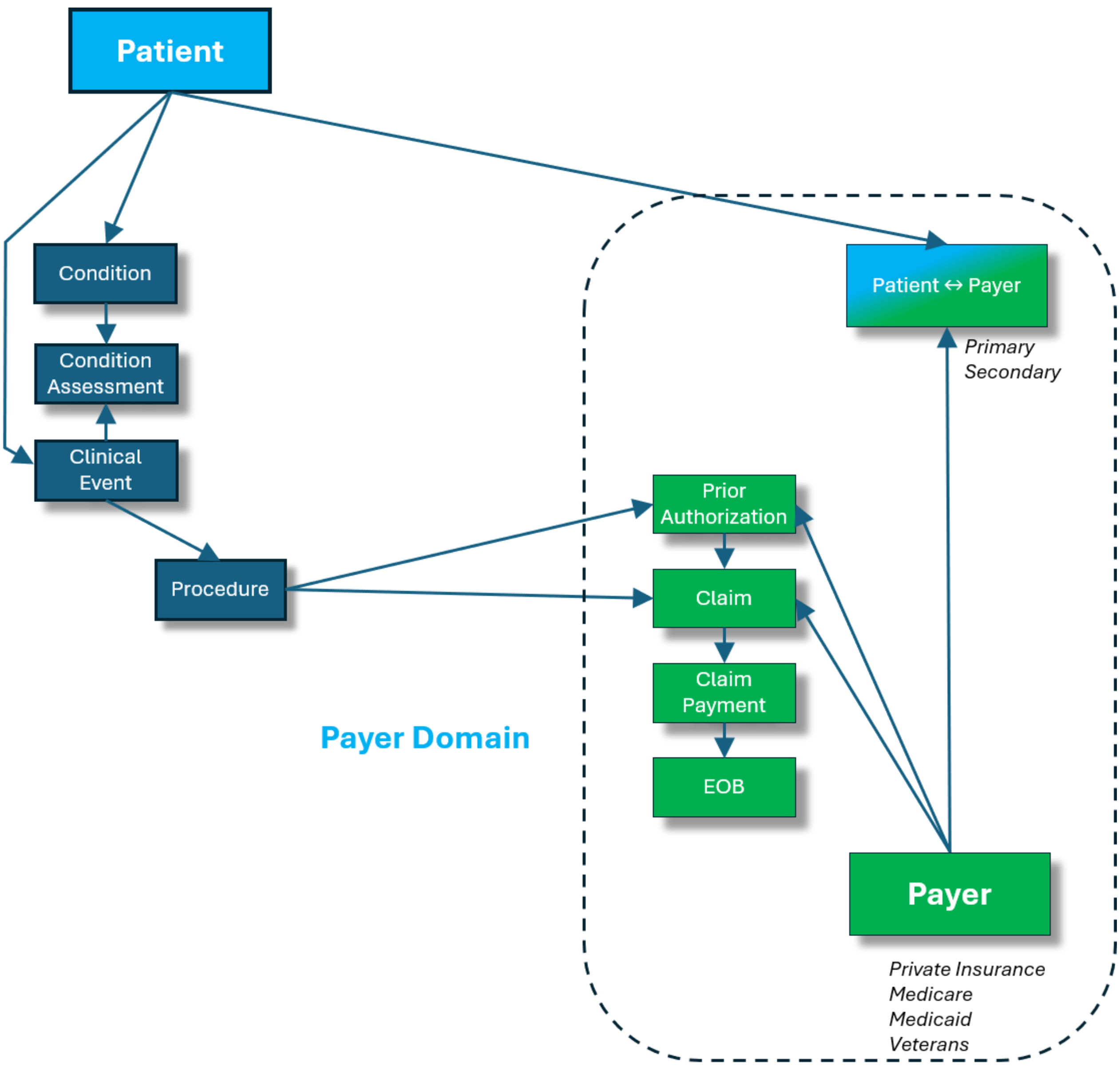
For example, a condition might first appear as a patient-reported medical history, later be evaluated by a primary-care physician, and subsequently be confirmed, questioned, or refined by a specialist. The longitudinal history should preserve each assessment, when it occurred, who made it, and the evidence supporting it, rather than replacing the history with only the most recent conclusion.

Clinical events may also be related to other clinical events. A primary-care visit may result in a referral to a specialist; a specialist consultation may lead to a procedure; and a procedure may result in subsequent follow-up visits. **Clinical Event Relationship** preserves these connections so that related encounters can be understood as part of the patient's continuing care rather than as isolated events.

Together, these relationships transform a collection of healthcare records into a longitudinal clinical history. The original clinical information remains available, while extracted and reconciled knowledge helps providers understand what conditions the patient has had, how those conditions have changed, what care has been provided, and what evidence supports those conclusions.

The Payer Domain

Healthcare also produces information describing how care is authorized, billed, and paid. The **Payer Domain** connects this administrative information to the same patient and clinical history used by providers.



Patient ↔ Payer identifies the organizations responsible for a patient's healthcare coverage. Because coverage changes over time and patients may have both primary and secondary coverage, these relationships must retain the periods during which each coverage relationship was valid.

Some procedures require **Prior Authorization** before they can be performed. After services are provided, **Claims** identify the procedures for which payment is requested. **Claim Payments** record how those claims were adjudicated and paid, while the **Explanation of Benefits (EOB)** communicates the disposition of the claim and the patient's financial responsibility.

Connecting this information to the clinical history provides capabilities that neither the clinical nor payer information can provide independently. Prior authorization can be supported by relevant clinical history rather than repeatedly assembling documentation from individual providers. Claims can be compared with the procedures and clinical information from which they originated, potentially reducing missing information, duplicate submissions, and manual review.

Over time, connecting clinical information with claims and payment information also makes it possible to examine the relationship between **care, cost, and outcomes**. Providers can better understand prior treatment and its results, patients can see a more complete history of the care they received, and payers can evaluate the cost and effectiveness of alternative treatments across comparable patient populations.

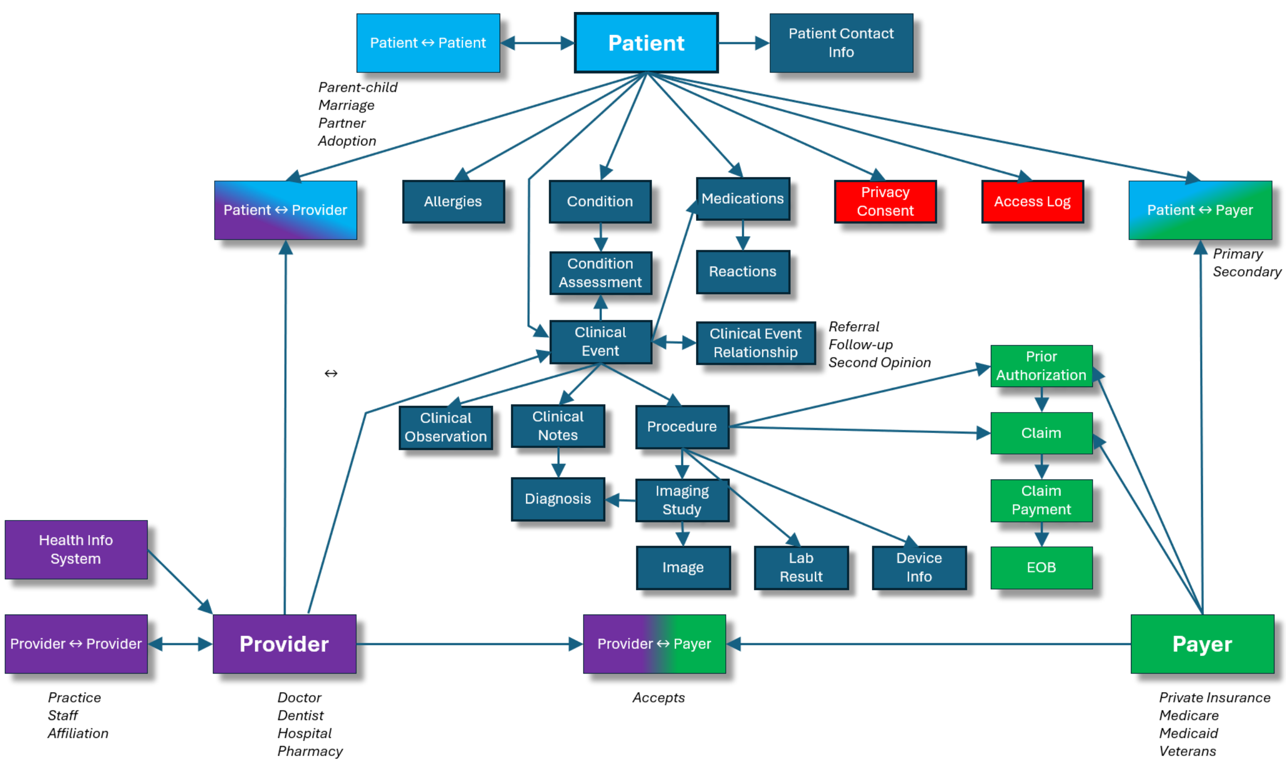
The objective is not to merge provider and payer operational systems. Each continues performing the functions for which it was designed. The longitudinal architecture provides the common patient context that allows clinical and administrative information distributed across those systems to be understood together.

Bringing the Information Together

The previous diagrams examined portions of the conceptual model separately so that their relationships could be seen more clearly. Together, they form a single longitudinal view of the patient.

The complete model connects the patient's conditions and clinical history with the providers who delivered care and the payers that authorized and paid for it. Information that today resides in separate Healthcare Information Systems and payer systems can therefore be related through the patient and the clinical events that occurred over time.

The value of the architecture does not come simply from collecting more data. It comes from preserving these relationships so that clinical information can be interpreted in context and used by providers, patients, and payers.



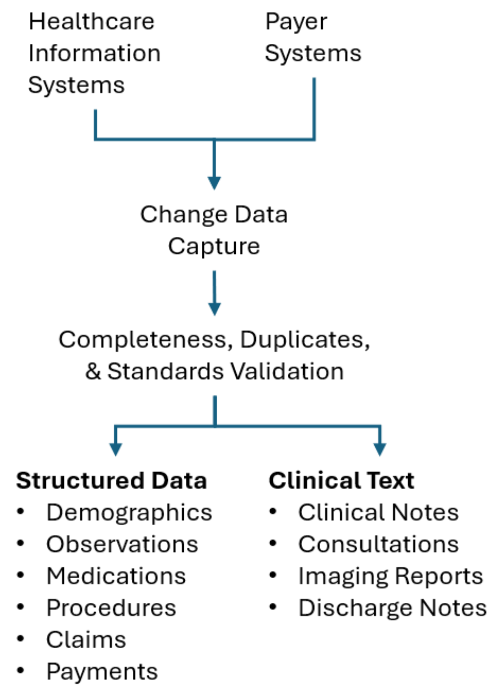
How the Longitudinal History Is Populated

The conceptual model describes the information that would be available in the longitudinal patient history. Populating and maintaining that history requires continuously collecting information from Healthcare Information Systems and payer systems and transforming it into a consistent longitudinal representation.

Not all this processing requires AI. Much of it uses well-established data integration techniques. AI becomes important when clinically significant knowledge must be extracted from narrative information or when information from multiple sources must be reconciled.

Receiving Complete and Reliable Data

The first step is conventional **Change Data Capture (CDC)**. Participating systems must identify information that has been added, changed, or deleted since the previous exchange.



The receiving architecture must determine whether expected data was received, detect duplicate transmissions, recognize corrections to previously received information, and prevent retransmitted information from being interpreted as new clinical activity. Persistent source identifiers are important so that a corrected clinical note, laboratory result, claim, or other record can be associated with the information it replaces or modifies.

This processing is mundane compared with AI, but it is essential. No subsequent analysis can produce a trustworthy longitudinal history if source information is missing, duplicated, or incorrectly interpreted as new.

Validating and Standardizing Structured Information

Much of the information received from healthcare and payer systems is already structured. Patient demographics, clinical observations, medications, procedures, laboratory results, claims, and claim payments contain defined data elements that can be validated and normalized upon receipt.

Healthcare interoperability standards substantially simplify this task by defining formats, identifiers, terminology, and coding systems. The longitudinal architecture must nevertheless validate incoming values, recognize unsupported or locally defined values, normalize differences where appropriate, and retain the original source representation.

Blood pressure, medication, procedure, or claim should not acquire a different meaning simply because it originated in a different Healthcare Information System.

Extracting Knowledge from Clinical Text

Some of the most important information about a patient is contained in narrative documents rather than structured columns. Clinical notes, discharge summaries, operative reports, specialist consultations, and imaging reports may describe conditions, symptoms, diagnoses, clinical observations, treatment decisions, treatment response, adverse effects, uncertainty, and historical information.

Large Language Models and other AI techniques can be used to identify this information and transform clinically significant statements into structured knowledge.

The original document should not be replaced by the extracted information. The complete Clinical Note remains part of the longitudinal history. Extracted knowledge retains a link to the source document—and, where practical, to the portion of the document supporting the extraction—so that a provider can verify how a conclusion was reached.

For example, a note might state that a patient's knee pain improved for several weeks following an injection but subsequently returned. Knowledge extraction could identify the condition under discussion, the procedure, the reported improvement, its duration, and any subsequent recurrence, while preserving the original statement as supporting evidence.

Identifying Redundant Information

Healthcare information is frequently repeated. A diagnosis may appear in a structured field, in a clinical note, in a specialist's report, and again in subsequent notes that repeat information from earlier encounters.

The longitudinal architecture should not interpret every occurrence as independent evidence.

Patient identity, provider, clinical event dates, source identifiers, terminology, and other contextual information can be used to determine whether apparently similar information represents a new clinical observation, a confirmation from an independent source, or a repetition of information already received.

The original source records remain available. The objective is not to delete duplication, but to prevent duplicated information from distorting the longitudinal knowledge derived from those records.

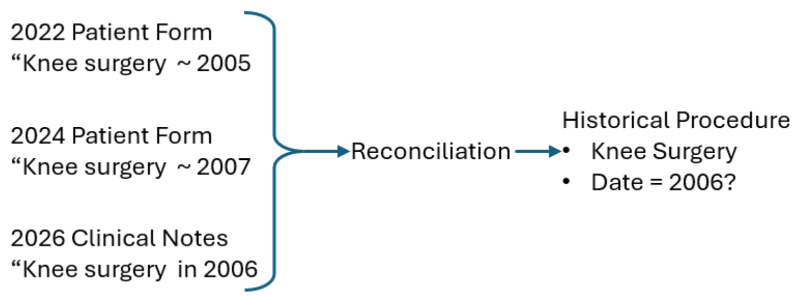
Reconciling Clinical History Over Time

A particularly important problem is determining how newly received information relates to the patient's existing Conditions and Condition Assessments.

A newly extracted diagnosis might identify a new condition. It might confirm or refine an existing condition. It might represent a historical condition reported by the patient. It might contradict an earlier assessment. It might also be uncertain or subsequently ruled out.

Patient-reported history presents a different reconciliation problem. Patients are frequently asked to provide previous medical conditions, surgeries, and approximate dates. Their recollection may change each time the information is requested. A surgery reported as occurring in 2005 on one questionnaire might be reported as occurring in 2007 several years later. These should not automatically be interpreted as two separate surgeries, nor should the later answer simply replace the earlier one.

Getting the Data Right



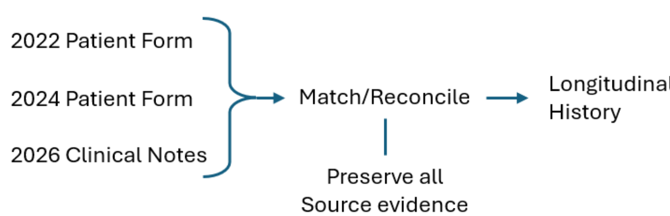
The longitudinal history should preserve what was reported, when it was reported, and the source of the information while attempting to determine whether the reports refer to the same historical event. If more authoritative evidence later becomes available—for example, the original surgical record—it can further refine the history without eliminating the earlier patient-reported information.

Preserving Time and Provenance

Healthcare information cannot be understood without knowing **when it was true and where it came from**.

The date a record was received is not necessarily the date of the clinical event. The date a provider documented a condition may differ from when the condition began. A clinical note written today may describe surgery performed twenty years ago. A medication may have been prescribed on one date, started later, changed several times, and eventually discontinued.

Getting the History Right



The longitudinal architecture therefore must preserve temporal context rather than reducing the patient's history to only the latest values.

It must also preserve provenance. Clinical knowledge should remain traceable to the provider, clinical event, source system, source document, and supporting evidence from which it was obtained.

Together, temporal context and provenance allow the longitudinal history to answer not merely **what is known about the patient**, but also **when it was known, who reported or assessed it, how that understanding changed, and what evidence supports it**.

From Data Collection to Clinical Knowledge

The result is more than a repository containing copies of healthcare records.

Structured information is validated and normalized. Narrative information is converted into evidence-linked clinical knowledge. Repeated information is recognized. Conflicting assessments are preserved and reconciled. Changes are maintained over time, and every significant conclusion remains traceable to its source.

That transformation is what allows information distributed across many Healthcare Information Systems and payer systems to become a usable longitudinal patient history.

Conclusion

A longitudinal patient history is more than a collection of records retrieved from different healthcare organizations. Its value comes from connecting information about patients, providers, clinical events, conditions, treatments, and payer activity while preserving when information was known, where it came from, and the evidence supporting it.

Much of this information already exists in Healthcare Information Systems and payer systems. Structured information can be validated and normalized, while AI can extract additional clinical knowledge from narrative documents. That information must then be matched, reconciled, and maintained over time without losing conflicting information or its provenance.

As discussed in the previous paper, existing Healthcare Information Systems and TEFCA do not provide a mechanism to notify a national longitudinal architecture whenever relevant information changes in another system. On-demand interoperability allows information to be retrieved, but maintaining a current longitudinal history requires a reliable way to identify new, changed, and deleted information as it occurs.

Creating that history is only part of the architecture. The next paper examines how relevant information can be retrieved from it, how AI can use that information to assist clinical decision-making, and how those capabilities can be made available to existing Healthcare Information Systems.

Acknowledgements
Microsoft Word was used for document preparation and spell checking. Grammarly was used to suggest improvements in clarity and wording. ChatGPT was used as a research and development aid to investigate topics, challenge assumptions, explore alternative approaches, refine and summarize text, and suggest concepts for illustrations. The author is responsible for the architecture, conclusions, content selection, and final editing.